

Computer Networks

Lecture 5: Connecting Devices

Content



- **TCP/IP Protocol Suite**
- **Addressing**
- **Connecting devices**

TCP/IP Protocol Suite

The Internet Protocol Suite (commonly known as TCP/IP) is the set of communications protocols used for the Internet **and other similar networks**.

It is named from two of the most important protocols in it:

- Transmission Control Protocol (TCP) and
- Internet Protocol (IP),

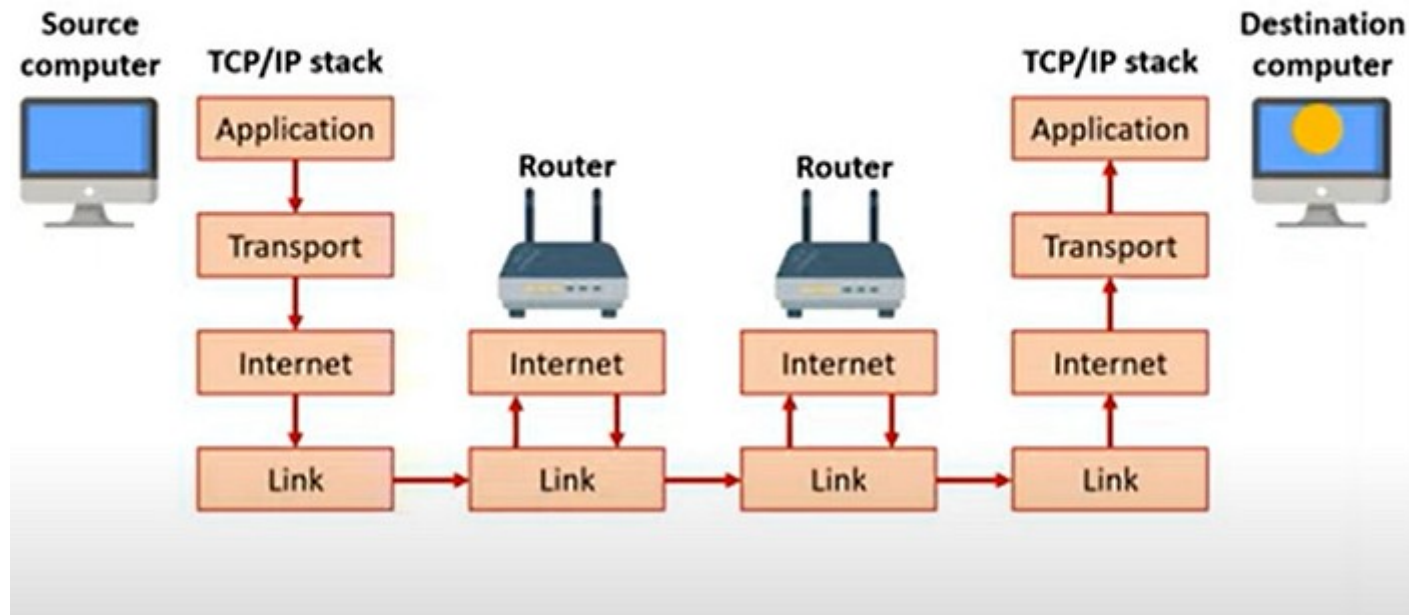
which were the first two networking protocols defined in this standard.

TCP/IP Protocol Suite

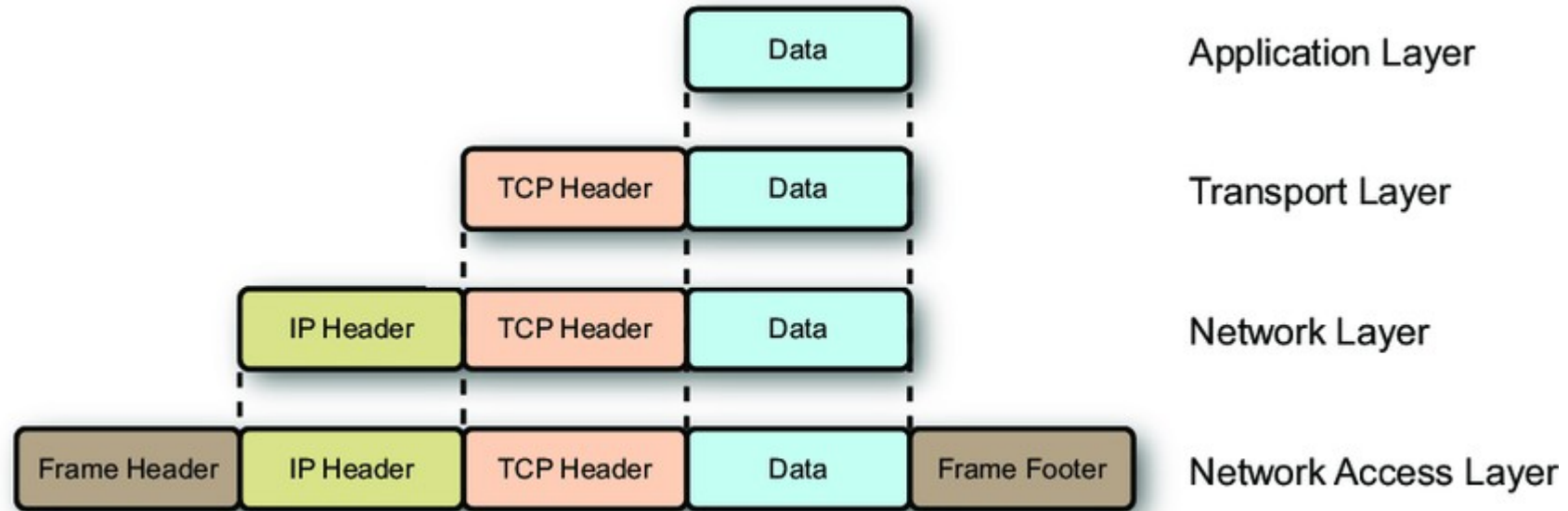
OSI	TCP/IP
Application Layer	Application Layer TELNET, FTP, SMTP, POP3, SNMP, NNTP, DNS, NIS, NFS, HTTP, ...
Presentation Layer	
Session Layer	
Transport Layer	Transport Layer TCP , UDP , ...
Network Layer	Internet Layer IP , ICMP, ARP, RARP, ...
Data Link Layer	host-to-network/ Link Layer FDDI, Ethernet, ISDN, X.25,...
Physical Layer	

TCP/IP Stack

TCP/IP protocol and the use of layers

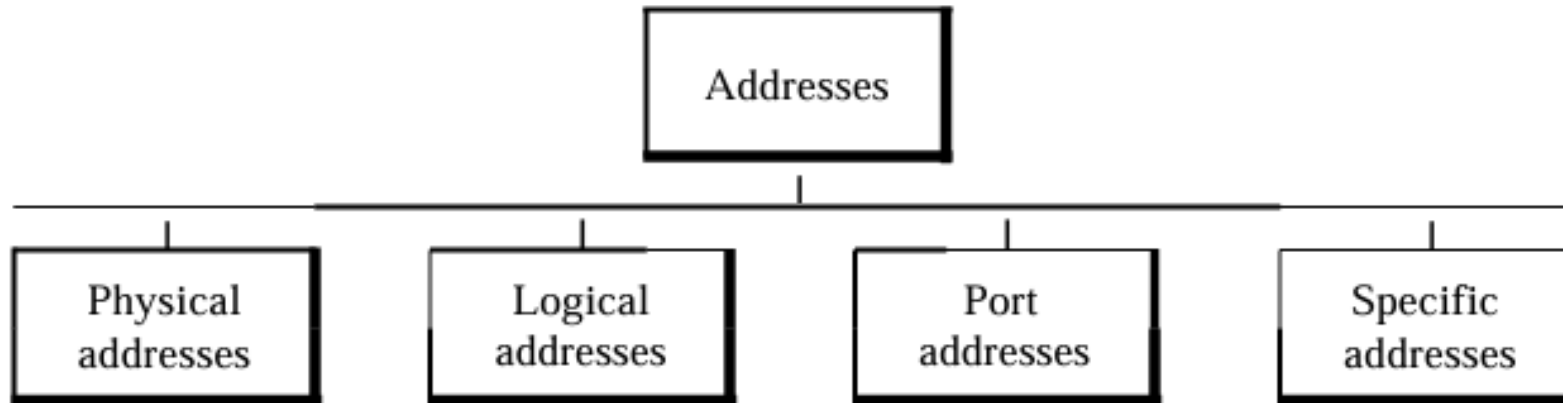


TCP/IP Data Encapsulation



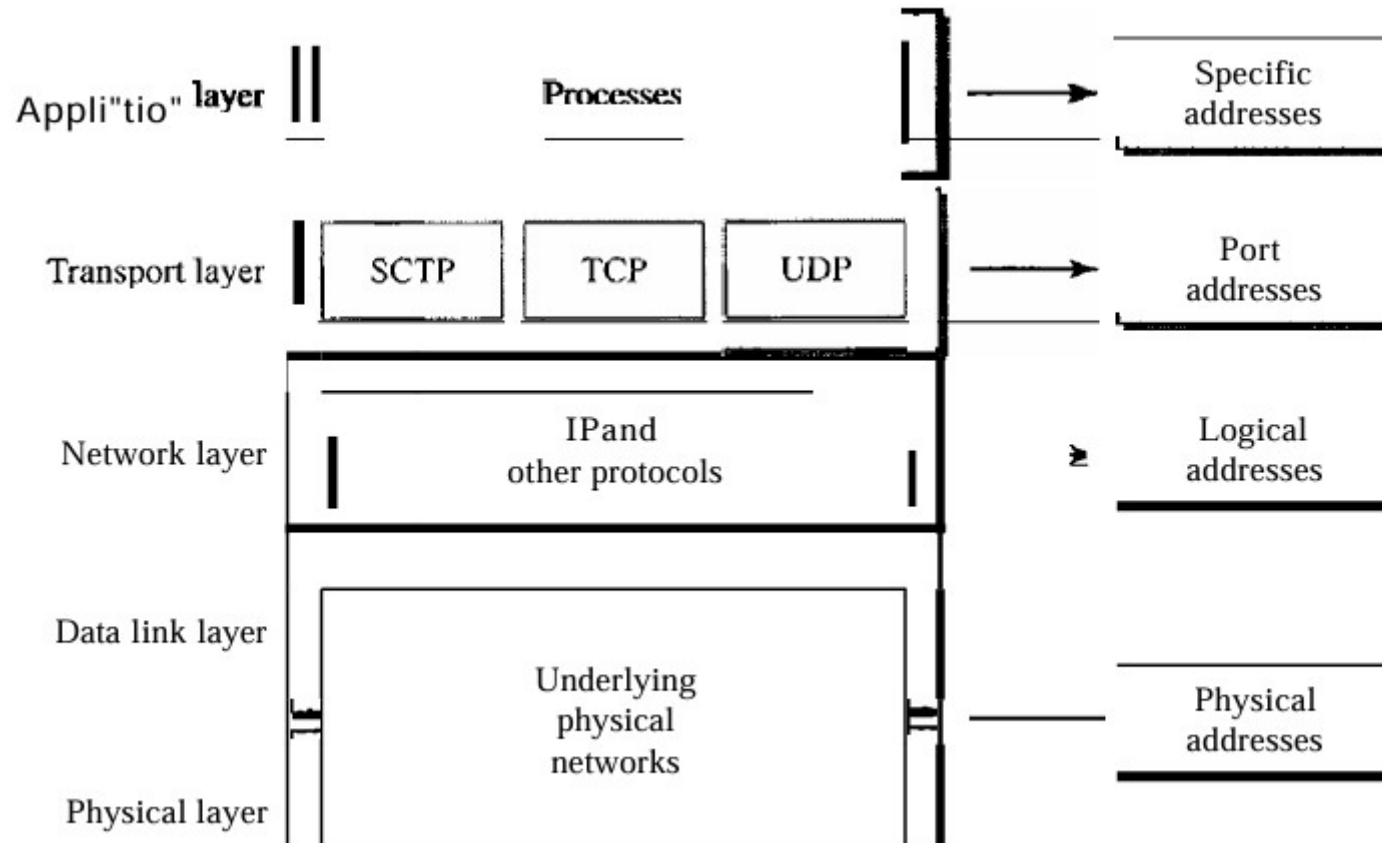
TCP/IP Data Encapsulation

Four levels of addresses are used in an internet employing the TCP/IP protocols:



TCP/IP Layers vs Addresses

Four levels of addresses are used in an internet employing the TCP/IP protocols:



TCP/IP Layers Addresses

Physical Addresses: It is the address of a node as defined by its LAN or WAN.

Logical Addresses: IP address (IPv4 or IPv6), is a hierarchical, software-assigned identifier used at the Network Layer (Layer 3) to uniquely route data across interconnected networks

Port Addresses: In the TCP/IP architecture, the label assigned to a process is called a port address.

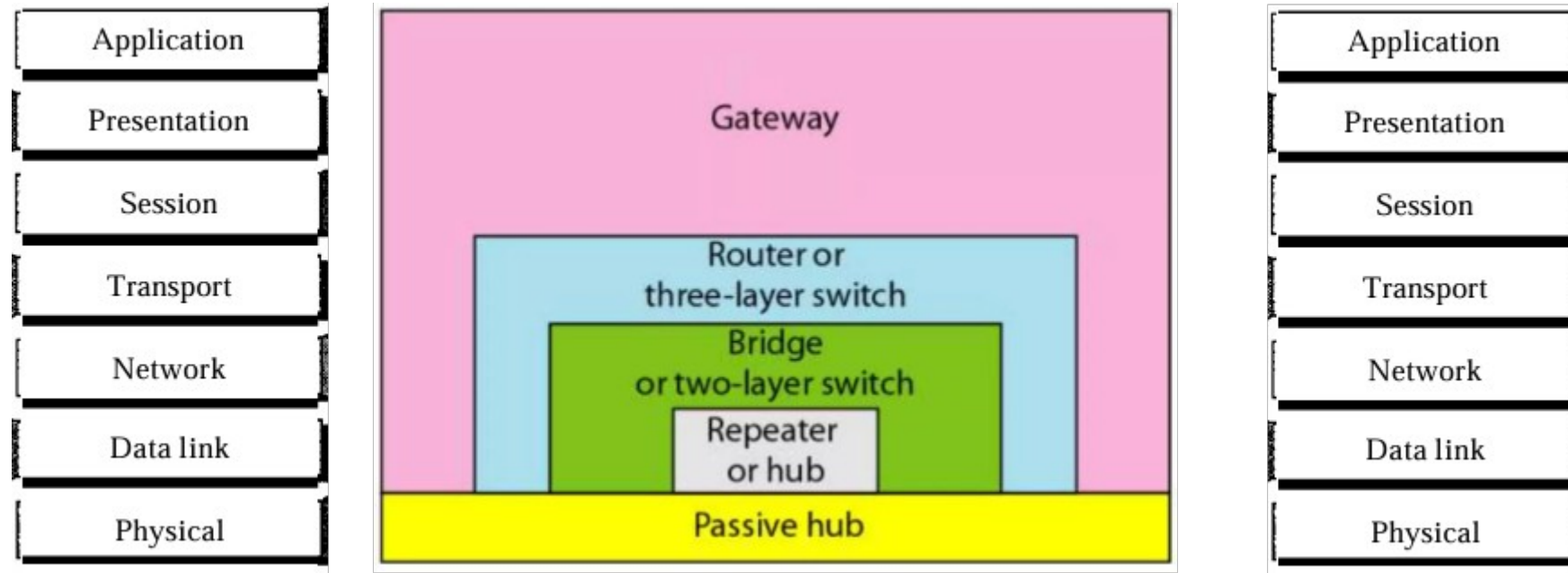
Specific Addresses: Some applications have user-friendly addresses that are designed for that specific address.

Connecting Devices

Networking devices are components used to connect computers or other electronic devices Together so that they can share files or resources like printers or fax machines. These are also called communicating devices.

- Repeater
- Hub
- Bridge
- Switch
- Router
- Gateway
- Brouter
- Modem

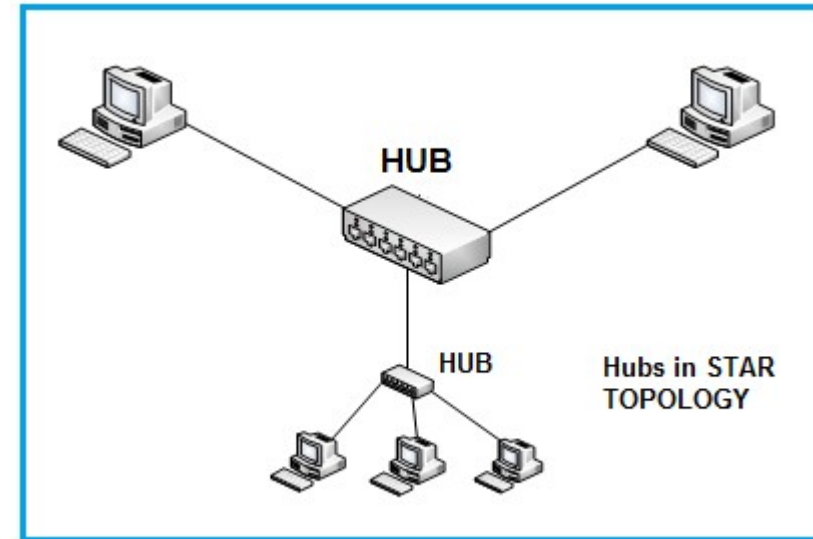
Connecting Devices



Passive Hub

A passive hub, or unpowered hub, acts as a basic, non-powered central connector in star-topology networks, simply bridging cables to pass signals between devices without amplification or regeneration

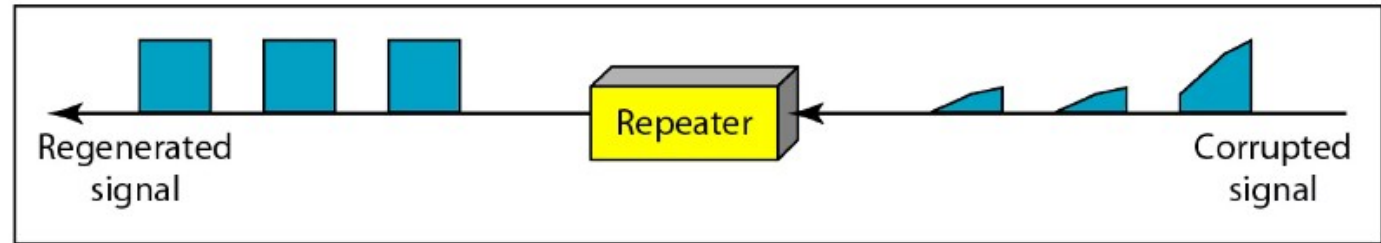
When a signal is received on one port, it is broadcasted to all other ports without any cleaning, processing, or strengthening.



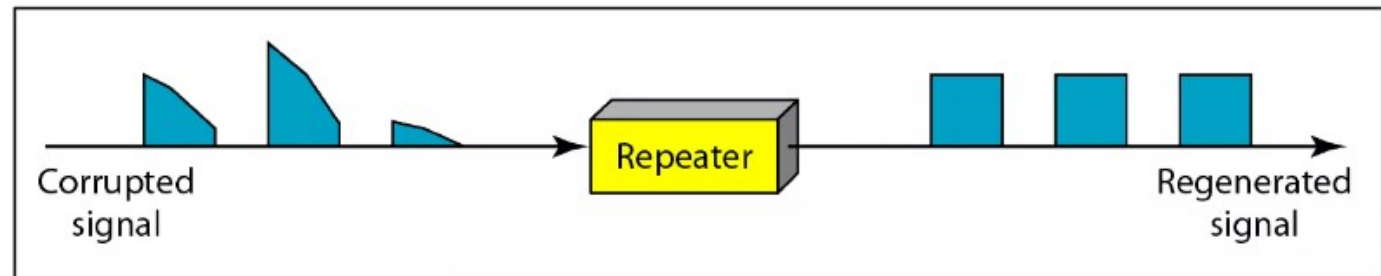
Repeater/Active Hub

A repeater is a regenerator not an amplifier

It is used to extend the transmission distance of a signal by regenerating or amplifying it before it becomes too weak or distorted.



a. Right-to-left transmission.



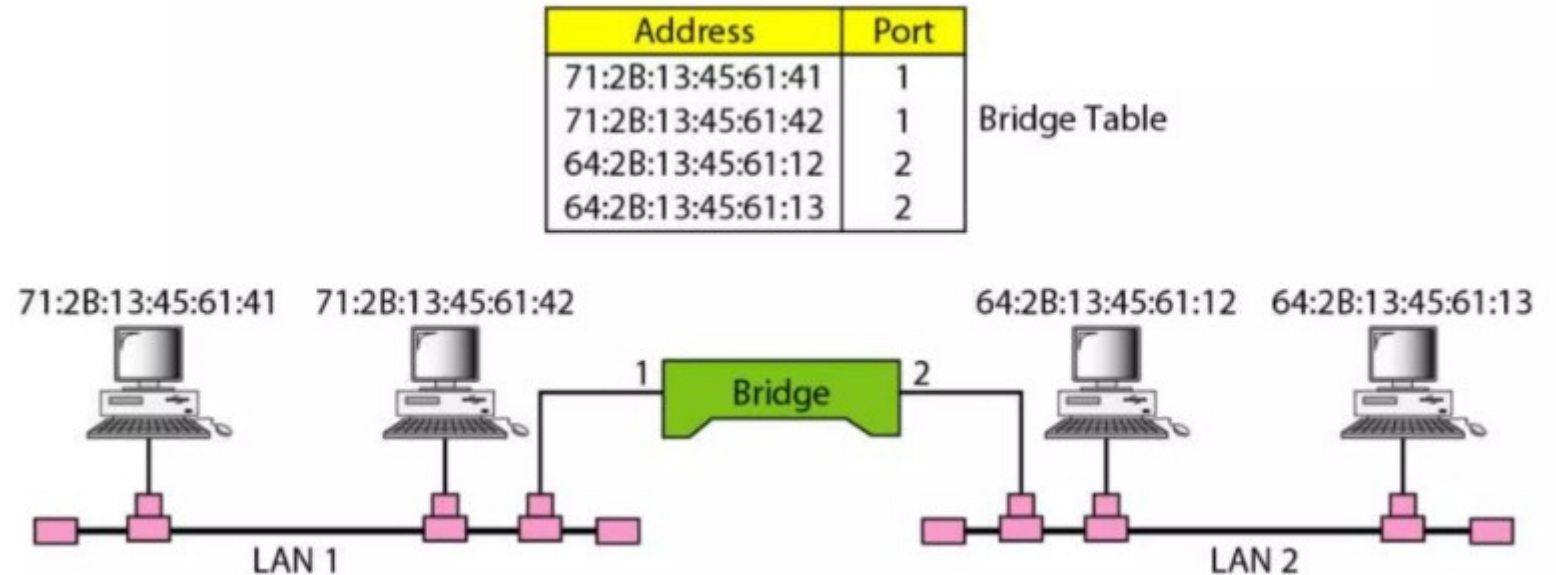
b. Left-to-right transmission.

A repeater will forward every frame it has, there is no filtering

Bridge

A bridge checks the MAC addresses. It has a table used in filtering decisions. This table shows the mapping of MAC addresses to the ports

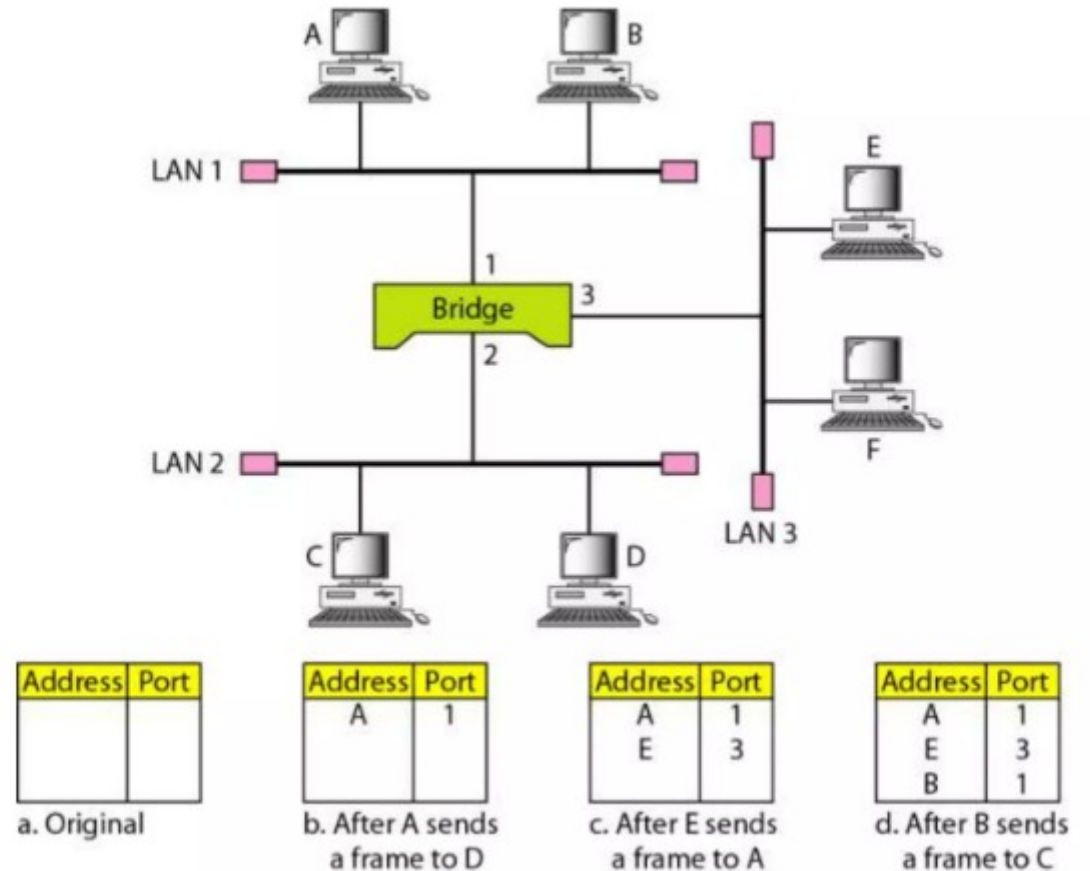
A bridge does not alter the MAC addresses in the frame.



Bridge

Process of Learning

- **Filter:** If the destination address is already mapped to the *same* port the frame arrived on, the bridge drops (filters) the frame because the recipient is on the same segment.
- **Forward:** If the destination is mapped to a *different* port, the bridge sends the frame only out of that specific port.
- **Flood:** If the destination is **unknown** (not in the table), the bridge "floods" the frame to all other ports except the incoming one.



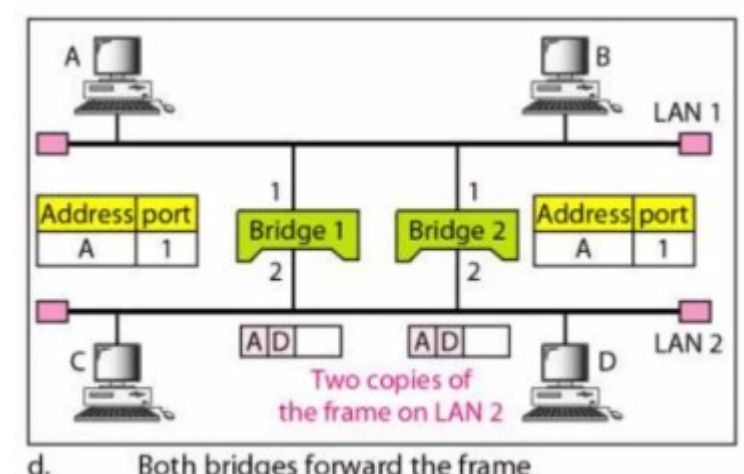
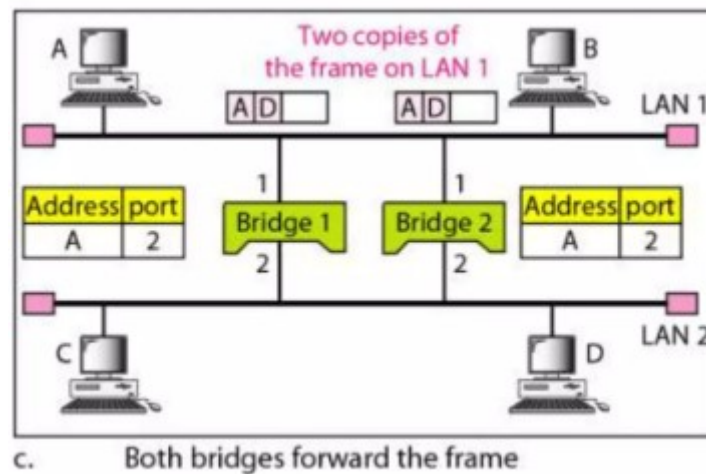
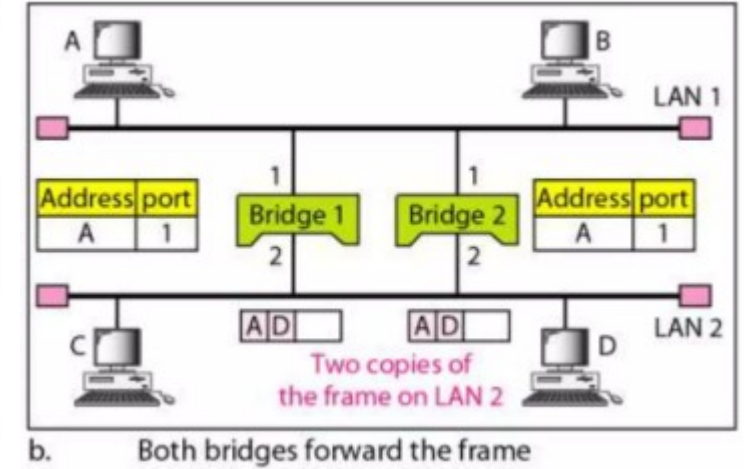
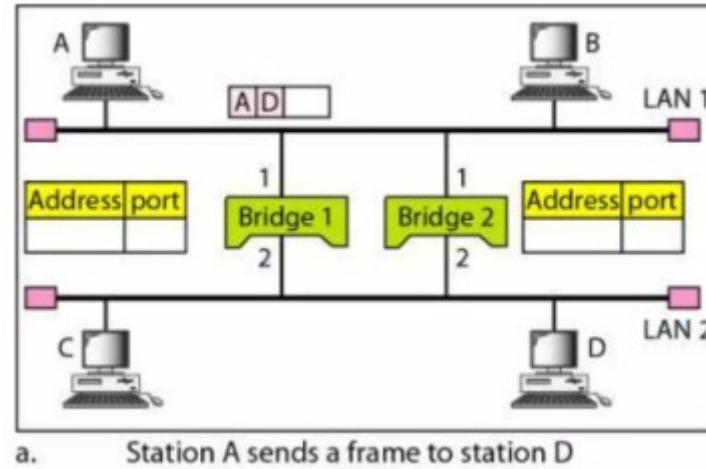
Loop Problem in Learning Bridge

A learning bridge forwards frames based on its MAC address table.

If loops exist in the network topology, a frame may circulate endlessly, causing:

- Multiple copies of the same frame
- Broadcast storms
- MAC table instability
- Network congestion or failure

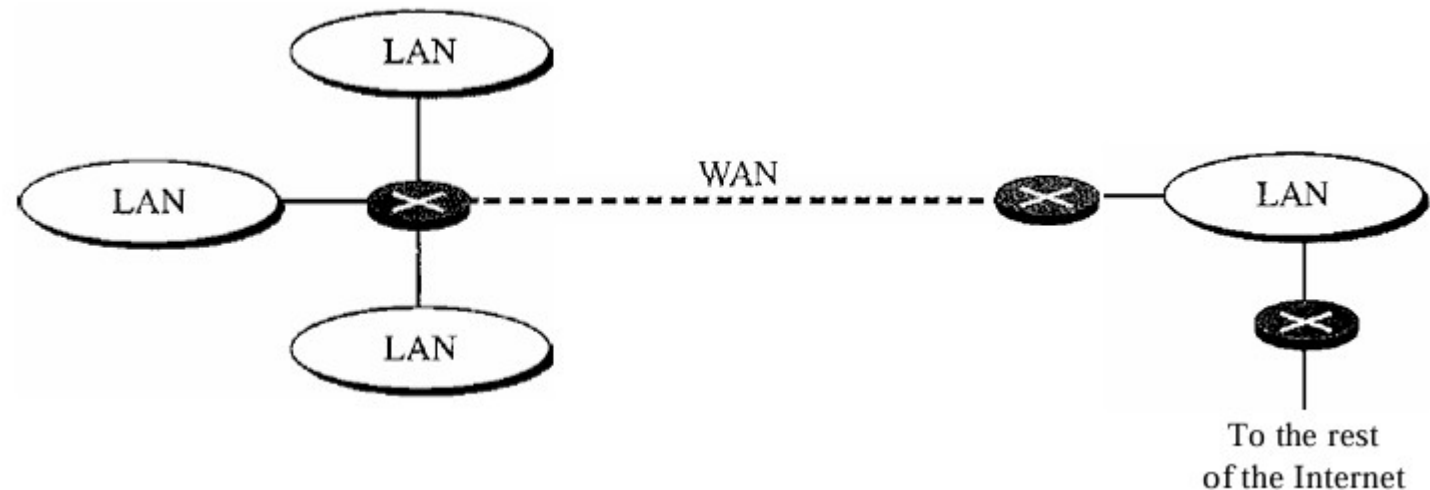
This happens because Ethernet frames do not have a TTL (time-to-live) field.



Router

A router is a three-layer device that routes packets based on their logical addresses (host-to-host addressing).

The routing tables are normally dynamic and are updated using routing protocols.

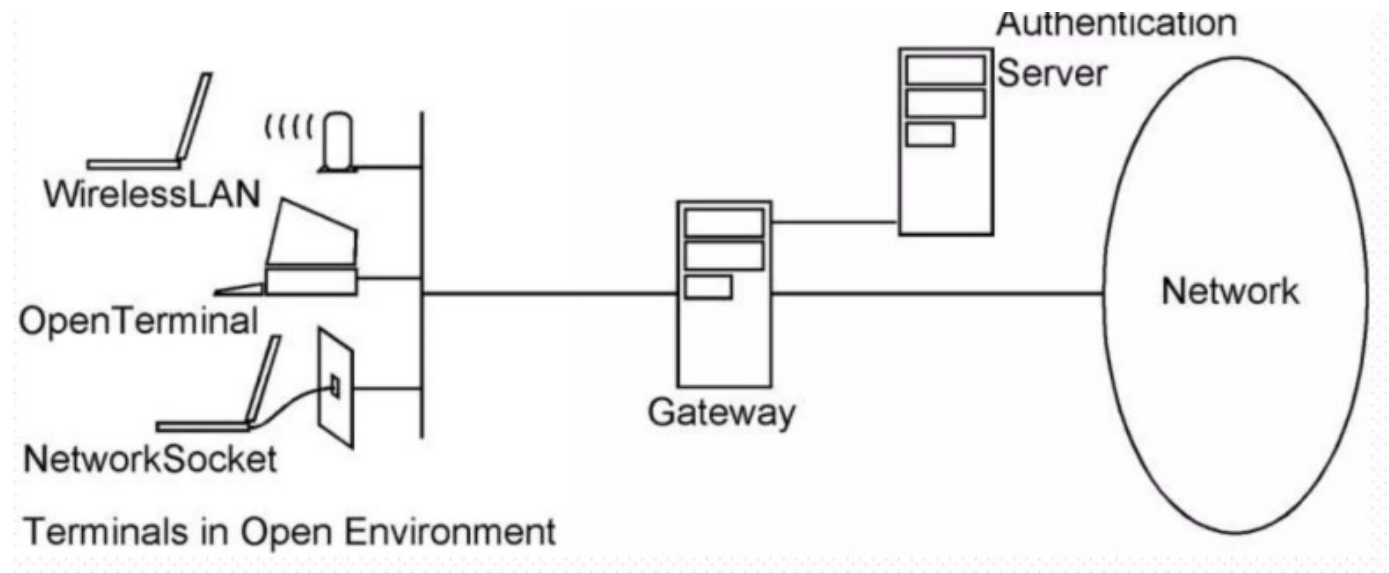


Gateway

A gateway is normally a computer that operates in all five layers of the Internet or seven layers of OSI model. A gateway takes an application message, reads it, and interprets it. This means that it can be used as a connecting device between two internetworks that use different models.

Example:

Email Gateway: Converts email protocols such as SMTP to POP3 or IMAP.



Brouter

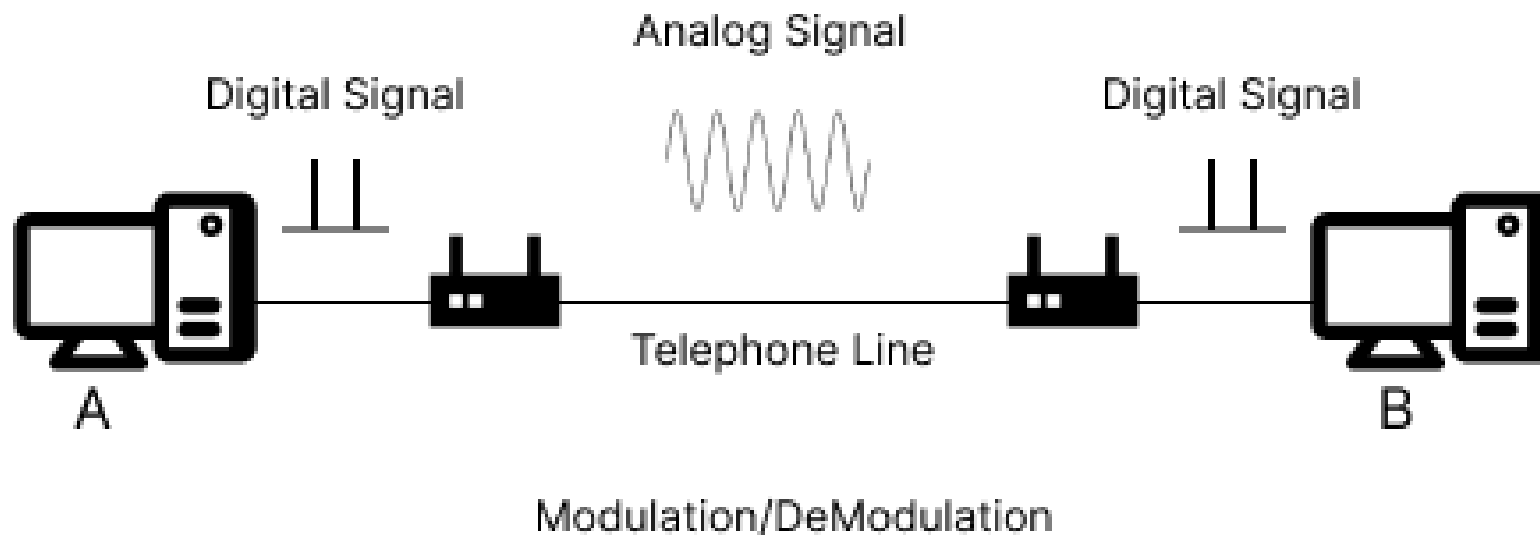
A brouter (bridge router) in a computer network combines features of a bridge (Layer 2) and a router (Layer 3)

A brouter operates at two different layers of the OSI model simultaneously:

- Routing (Layer 3): For "routable" protocols like TCP/IP, the device acts as a router, directing traffic between different networks using IP addresses.
- Bridging (Layer 2): For "non-routable" protocols (like some legacy traffic), it acts as a bridge, forwarding frames between segments of the same network based on MAC addresses.

Modem

A modem (modulator-demodulator) is a critical networking device that enables internet access by converting digital data from devices (computers, routers) into analog signals for transmission over ISP lines (cable, telephone, fiber), and vice versa.



Question

1. Two computers are connected using a device. When one computer sends data, all other connected computers receive the signal, but only the intended one processes it. What type of connecting device is most likely used?
2. In a small office network, replacing the existing connecting device reduces unnecessary network traffic and improves data security, even though the number of computers remains the same. Which type of connecting device could have been replaced?
3. A university connects its internal network to the Internet using a device that:
 1. Chooses the best path for data
 2. Can connect different networks
 3. Uses logical addressing instead of physical addressingIdentify the type of connecting device and justify your answer logically.